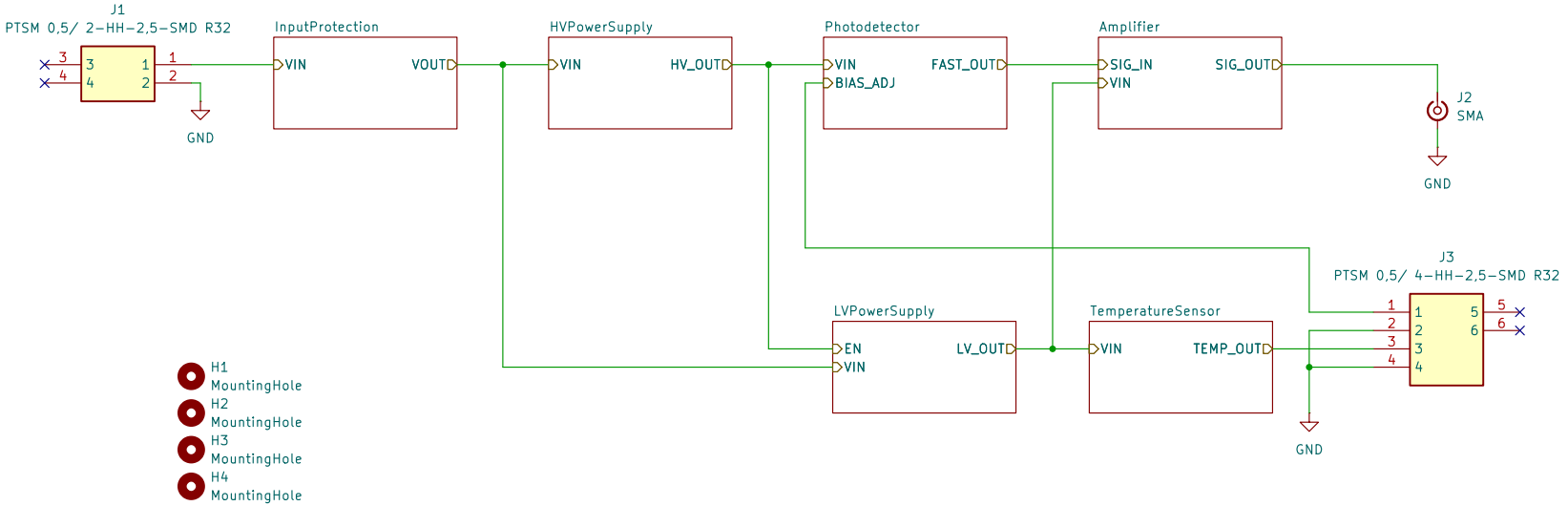


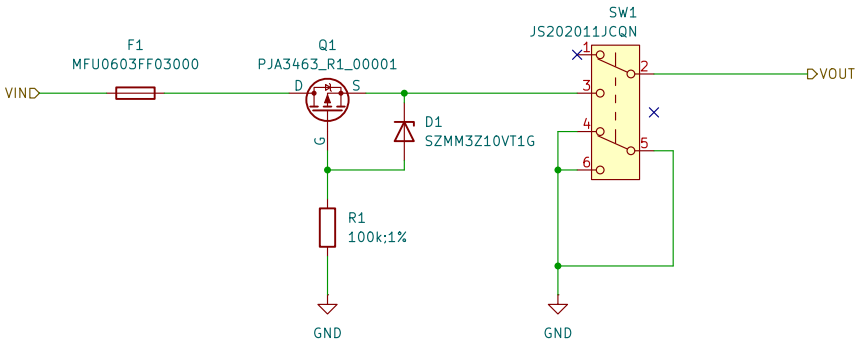

**Hannes Duske**

Sheet: /
File: SinglePhotonDetector.kicad_sch

***Title: Single Photon Detector***

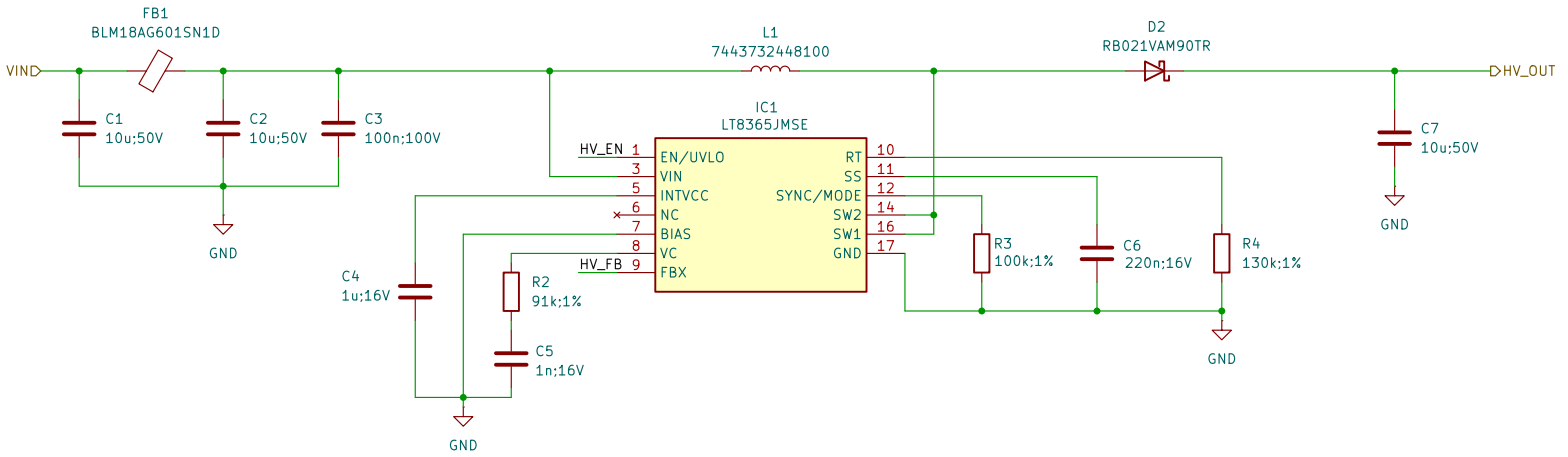
| | | |
|---|---|---|
| Size: A4 | Date: 2026-04-08 | **Rev: 0.1** |
| KiCad E.D.A. 10.0.6 | | Id: 1/7 |

# Input Protection

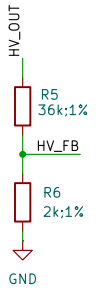



| | | |
|---|---|---|
| Sheet: /InputProtection/ | | |
| File: InputProtection.kicad_sch | | |
| **Title:** | | |
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.6 | | Id: 2/7 |

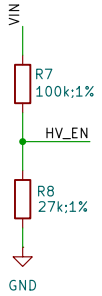
# HV Power Supply




## Feedback Network




## Enable Network

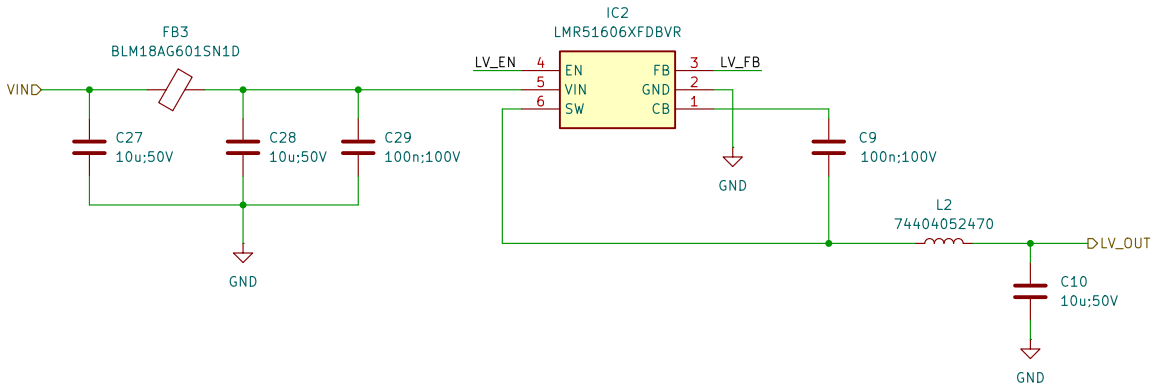



Undervoltage shutdown threshold is 7.5 V with the selected enbale network resistors.

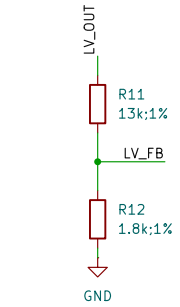
Output Voltage is 30.4V with the selected feddback network resistors.

| | | |
|---|---|---|
| Sheet: /HVPowerSupply/<br>File: HVPowerSupply.kicad_sch | | |
| **Title:** | | |
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.6 | | Id: 3/7 |

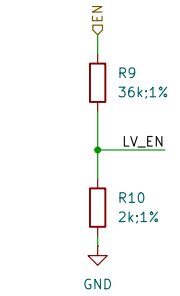
# LV Power Supply




## Feedback Network




## Enable Network

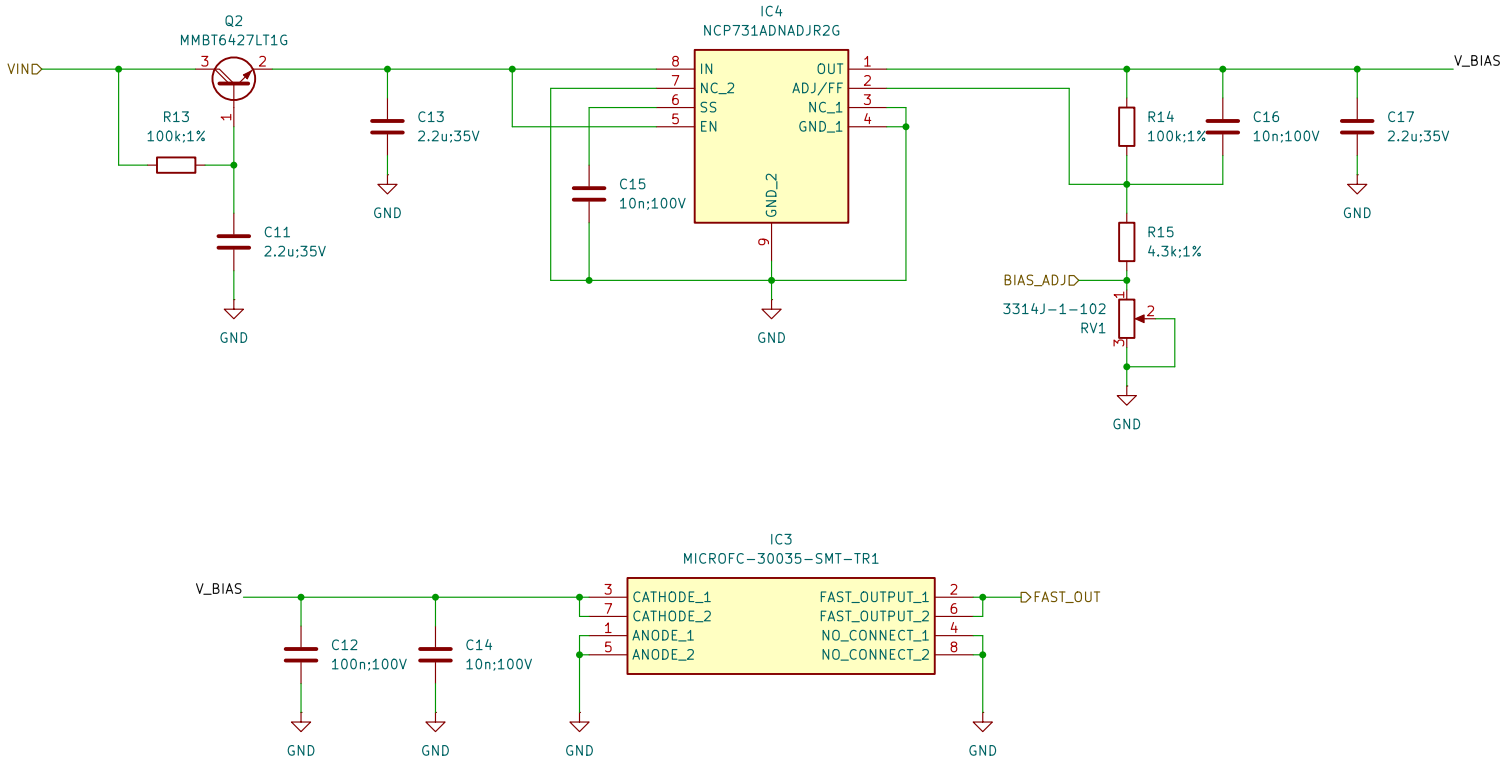



Enable voltage threshold is 23.3 V
so the low voltage supply only starts
when the HV supply is stable.

Output voltage is 6.6 V with the selected
feedback network resistors.

| | | |
|---|---|---|
| Sheet: /LVPowerSupply/<br>File: LVPowerSupply.kicad_sch | | |
| **Title:** | | |
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.6 | | Id: 4/7 |

# Photodetector



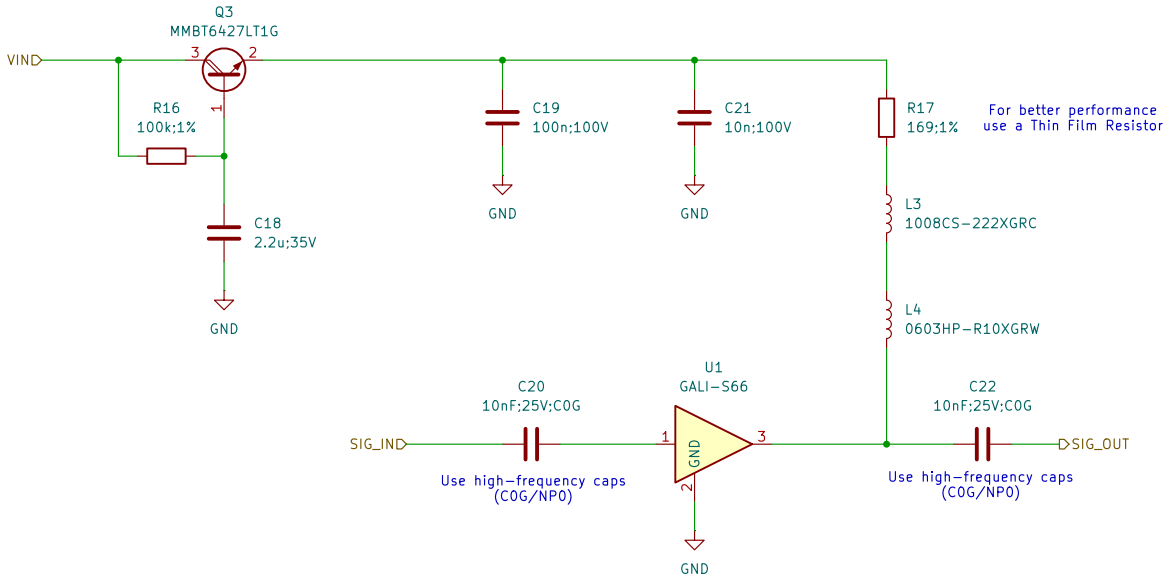




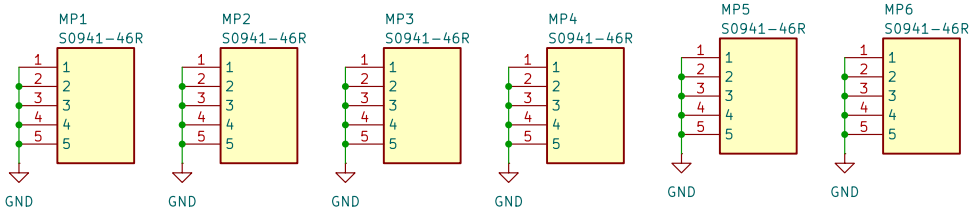
V_BIAS adjustable between
23.8 V ... 29.1 V with the
selected resistor values

| | | |
|---|---|---|
| Sheet: /Photodetector/<br>File: Photodetector.kicad_sch | | |
| **Title:** | | |
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.6 | | Id: 5/7 |

# Amplifier

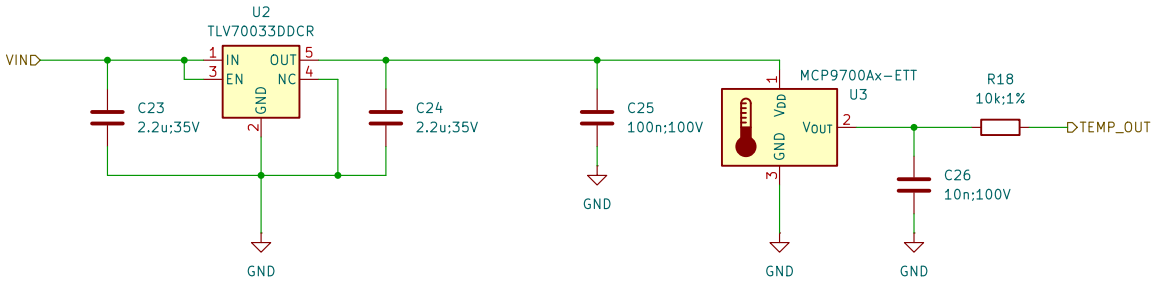



## Shielding Can Clips




| | | |
|---|---|---|
| Sheet: /Amplifier/ | | |
| File: Amplifier.kicad_sch | | |
| **Title:** | | |
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.6 | | Id: 6/7 |

# Temperature Sensor




| | | |
|---|---|---|
| Sheet: /TemperatureSensor/ | | |
| File: TemperatureSensor.kicad_sch | | |
| **Title:** | | |
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.6 | | Id: 7/7 |